Contents

The Unsupervised Learning Landscape 11

Geometry of Data and Distance 13

Representative-Based Clustering 15

Hierarchical Clustering 17

Density-based and Graph-based Clustering 19

Association Rule Learning 21

Linear Manifolds and Matrix Factorization 23

Non-Linear Dimensionality Reduction 25

Unsupervised Pretraining 27

Masked Modeling 29

Variational Inference 31

Diffusion 33

Zero-Shot and Few-Shot Learning 35

Domain Adaptation 37

OOD and Anomaly Detection 39

Evaluation Structure without labels 41

Weak Supervision 43

Data Programming and Label Modeling 45

Human-in-the-loop Learning 47

Imitation Learning 49

Boosting Strategies 51

Self-Training Loops 53

Model-in-the-loop Learning 55

Knowledge Distillation 57

Reinforcement Learning 59

Model-Free RL 61

Model-Based RL and World Models 63

Exploration 65

Index 67